Optimizing Tile Procurement to Reduce Wastage in Residential Construction: A Study on Operational Efficiency

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1. Executive Summary

Optimizing Tile Procurement to Reduce Wastage in Residential Construction: A Study on Operational Efficiency

Protech Group, a mid-sized residential construction firm based in Guwahati, Assam, faces recurring challenges in its tile procurement process. Despite consistent growth, material breakage and reordering inefficiencies lead to cost overruns and schedule delays during project finishing phases. The company currently applies a standard 8% procurement buffer (5% for breakage, 3% for sagging and future maintenance), yet emergency reorders remain frequent.

This project analyzes the firm's tile procurement workflow using primary data from 26 purchase orders. The methodology combined descriptive analytics in Google Sheets with stakeholder consultation to validate findings. Orders were categorized into planned phases versus emergency shortages. Actual wastage percentages were calculated for high-risk tiles, and buffer adequacy was simulated at different levels.

The analysis reveals that the standard 8% buffer is systematically insufficient for specific tile types. Balcony tiles (**111 RISER**) exhibited consistent wastage of 12-14% across project phases, forcing emergency reorders. A catastrophic 53% reorder for design tiles (**CARVED STELLA**) was traced to post-installation design changes – a process failure beyond inventory control. The true cost of these failure extends beyond material costs to include management overhead, workflow disruption, and value destruction.

Based on these findings, a comprehensive set of recommendations is proposed: a tiered buffer policy (8% for standard tiles, 15% for high-breakage tiles), an integrated design approval design with physical mock-up facilities, central maintenance stock, supplier collaboration for emergency buffer stock, and a procurement dashboard for early warning. These solutions address both inventory and process gaps, enhancing operational efficiency and project predictability.

2. Introduction

2.1 Organization Background

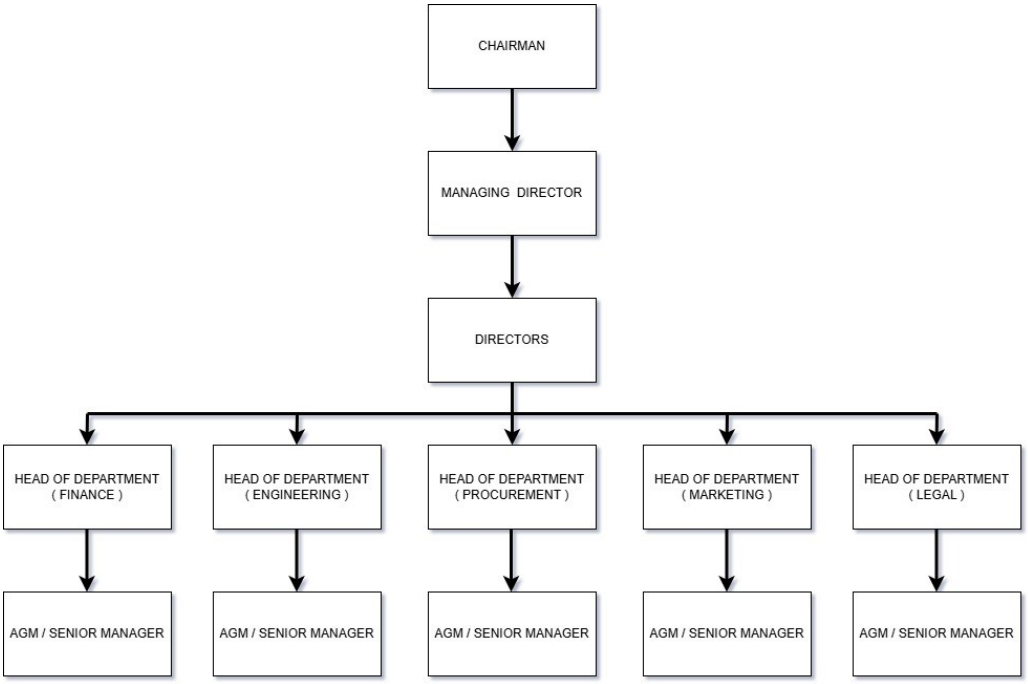
Protech Group is a prominent real estate developer based in Guwahati, Assam, with a history in the regional construction industry dating back to its establishment in 2005. Over the course of two decades, the company has developed a diverse portfolio that includes residential, commercial, and warehousing project. To date, the firm has completed more

than one million square feet of built-up area and delivered over 1600 individual units, establishing a consistent presence in the Northeastern Indian real estate market.

The company operates primarily within the B2C segment, focusing on the demand for urban housing solutions in Assam. Key developments, such as Protech Regency, Protech Pride, and Protech Tarun Nagar, serve as examples of the firm's approach to modern residential planning. The group's operational standards are centered on quality workmanship and a commitment to meeting project delivery schedules, which are core components of its service delivery to stakeholders.

Under the leadership of Managing Director Manoj Jalan, Protech Group maintains an annual turnover of approximately ₹200 crore. The company's growth is reflected in its reach, having provided homes to over 1250 families since its inception. By adhering to principles of structural reliability and sustainable construction practices, Protech Group continues to be a steady contributor to the ongoing urban and economic development of Guwahati.

Organization Chart:



2.2 Problem Statement

1. **Breakage Beyond Planned Buffer:** Despite ordering with a 5% buffer, tile breakage due to mishandling during unloading, shifting, and fitting often exceeds 8–12%, leading to material shortages.

2. **Inefficient Reordering and Workflow Disruption:** Emergency reorders are subject to long and uncertain transport lead times. While labor can be temporarily reassigned to other tasks, the unpredictability of material arrival creates significant management overhead and stress. Site managers and procurement teams must engage in reactive firefighting – coordinating with suppliers, expediting shipments, and rescheduling work – which diverts their focus from planned activities and increases the risk of cascading delays.

3. **Cost Overruns and Margin Pressure:** Rework, unplanned freight, and the hidden costs of management time and project uncertainty erode profitability on fixed-budget projects.

2.3 Project Objectives

- To analyze the firm's tile procurement workflow using primary data from purchase orders.
- To quantify actual wastage levels and reorder dependencies for different tile categories.
- To identify the root causes of buffer insufficiency through data analysis and stakeholder consultation.
- To simulate the adequacy of different buffer levels and determine optimal buffers for high-risk tiles.
- To recommend data-backed improvements in buffer planning, handling practices, design approval processes, and procurement scheduling.

3. Data Collection and Proof of Originality

3.1 Data Source and Collection Process

Meetings and Discussions:

Initial meetings were held with the Head of Department (HOD) at Protech Group to understand the business, its procurement workflow, and the challenges faced. Over three weeks, multiple discussions were conducted to identify the scope of the problem and the data available. The HOD provided access to purchase order records from the company's latest project, which was nearing completion. The data was shared in digital format (PDF files).

Field Notes Summary:

- **Meeting 1 (Introductory):** Understood business operations, tile procurement process, and the general challenge of reorders. HOD mentioned the 8% buffer and that breakage often exceeds it.
- **Meeting 2 (Data Exploration):** HOD shared purchase orders in PDF format. Discussed what data was available and what was missing. Agreed on scope of data collection.
- **Meeting 3 (Preliminary Findings):** Shared initial structured data with HOD for validation. Discussed anomalies and potential causes.
- **Meeting 4 (Root Cause Validation):** Presented preliminary analysis of 111 RISER and CARVED STELLA. HOD explained design change incidents and confirmed emergency reorder classification.

Data Collection:
The data was collected manually from 26 purchase order documents spanning the project duration. Each PO contained details such as order date, tile description, quantities (boxes and square feet), rate, supplier, and delivery terms.

Narrowing Down the Problem:
Initial discussions revealed that despite a standard 8% buffer, the company frequently faced reorders. The HOD specifically mentioned that breakage often exceeded 8-12%. This become the focal point of the analysis. Later discussions revealed that design changes were also a significant contributor.

3.2 Proof of Originality

a) Video Interaction with HOD of Procurement:

A video interaction was conducted with the HOD of Procurement, of Protech Group. The discussion covered:

- A brief introduction to the business and his role.
- Tile procurement process.
- The problem of tile breakage and reorders.
- Management of labour.

Google Drive Link:

https://drive.google.com/drive/folders/1jGgUbpAR9VivmqLReHJPdHmp__FSvaKu?usp=drive_link

b) Photographs:

Image 1: Sample Purchase Order Document

PURCHASE ORDER

protech Invoice To **Aero City Buildcon Pvt Ltd** Room No. 609, 6th Floor Protech Centre, Ganeshguri G.S Road Guwahati GSTIN/UIN: 18AALCA7758D1ZD State Name : Assam, Code : 18 E-Mail : info@protechgroup.in	Voucher No.	Dated **13-Jun-25**
		Mode/Terms of Payment **Within 15 Days**
	Reference No. & Date.	Other References **Local Freight Extra at Actual**
Supplier (Bill from) **CERAMIC ZONE** NH-37, DPS ROAD , KATABARI GUWAHATI KAMRUP METROPOLITAN GSTIN/UIN : State Name : Assam, Code : 18	Dispatched through	Destination **Garal Dharapur**
	Terms of Delivery **Ready Stock** **Site : Protech Dharapur** **Site Person :**	

Sl No.	Description of Goods	Due on	Quantity	Rate	per	Amount
1	**PD/T/S/011 Balcony Tiles&** *Model No : 111 Riser* *Size : 300x300 mm* *Thikness : 8.5 mm* *Approx Weight : 12.50 mm* *Sqft Per Box : 7.75* *Packing Box : 8 Pcs* *Total Box : 91 Box*	*13-Jun-25*	**705.2500 sqft.**	27.99	sqft.	**19,739.95**
	INPUT CGST					**1,776.60**
	INPUT SGST					**1,776.60**
	Less : ***Round Off***					**(-)0.15**
	Total		**705.2500 sqft.**			**Rs. 23,293.00**

Amount Chargeable (in words) *E. & O.E*

Indian Rupees Twenty Three Thousand Two Hundred Ninety Three Only

Remarks:
Prepared by Dipak Dhar
Company's PAN : .

Declaration
We declare that this invoice shows the actual price of the goods described and that all particulars are true and correct.

for Aero City Buildcon Pvt Ltd

Authorised Signatory

This is a Computer Generated Document

Source: Protech Group. Sensitive information has been erased for confidentiality. Full documents are available in the Google Drive link.

Image 2: Tiles at Site

Image 3: Tile Waste

c) Letter from the Firm:

A consent letter from Protech Group, signed by the Managing Director Mr. Manoj Jalan, confirming the authenticity of the data and the project collaboration, is provided below. The letter mentions that the student held multiple meetings with the Head of Department, who is responsible for procurement.

Google Drive Link:

https://drive.google.com/drive/folders/1vMcOvB5LkPgIQ5zySmTyeoKGF3tua4JP?usp=drive_link

d) Google Drive Link to All Original Materials:

All raw data, including the complete set of 26 purchase order documents, photographs, video, and the signed letter, are available at:

Google Drive Link:

https://drive.google.com/drive/folders/1qg5scFs3DZ6VIiMGzE4uoAaHcsH_ang6?usp=drive_link

The drive folder contains:

- All 26 purchase order documents
- The master Google Sheets file, Analysis worksheet
- Video recording
- Photographs
- Signed consent letter

3.3 Metadata and Data Structure

Field Name	Data Type	Description
Voucher No	Text	Unique purchase order identifier
Order Date	Date	Date when the order was placed
Project Site	Text	Construction site where tiles were delivered
Internal Code	Text	Company's internal code for

		tile series
Tile Type	Text	Type of tile (Balcony, Floor, Wall, etc.)
Model No	Text	Manufacturer's model/design name
Tile Size (mm)	Numeric	Tile dimensions (width*length) in mm
Thickness (mm)	Numeric	Thickness of the tile in mm
Pcs per box	Numeric	Number of pieces per box
Sqft per box	Numeric	Square feet coverage per box
Total boxes	Numeric	Total number of boxes ordered
Quantity ordered sqft	Numeric	Total square feet ordered
Rate per sqft	Numeric	Cost per square feet
Supplier	Text	Name of the supplier

Table 1: Metadata – Master Dataset Structure

4. Data Cleaning and Preprocessing

The raw data from the PDF files was transcribed into a structured Excel workbook. Key steps included:

1. Parsing the "Description of Goods":

Extracted internal tile codes, model names, sizes, thickness, packaging details (pieces per box, square feet per box), and total boxes.

2. Handling Missing Values:

- Missing **Thickness (mm)** (10 rows) and **Pcs per box** (4 rows) were imputed using rule-based assumptions derived from tile size, type, and industry standards (e.g., all 800*1600 mm floor tiles assumed to have 2 pieces per box; balcony tiles of 300*300 mm assumed 8 pieces per box).
- One missing Sqft per box value was calculated using the formula = **Quantity ordered sqft** / **Total boxes**.

3. **Derived Columns:**

- **Order Sequence:** Assigned by sorting data by **Internal Code**, **Tile Type**, **Model No**, **Project Site**, and **Order Date** (oldest to newest).
- **Is Reorder:** =IF(Order Sequence>1, "Y", "N").
- **Order Purpose:** Classified after analysis as **Planned** or **Emergency** based on quantity ratios and time gaps.

4. **Duplicate Removal:**

One duplicate order for **GRAIN WOOD WEAT** was identified and removed, resulting in a final dataset of 26 purchase orders.

5. **Data Validation:**

All entries were cross-checked with the HOD for accuracy, especially for tiles with multiple orders and unusual patterns.

5. Descriptive Statistics

The dataset comprises 26 purchase orders covering 16 unique tile models. Table 2 presents summary statistics for the core numerical variables across all orders, providing a high-level overview of procurement activity.

Statistic	Quantity Ordered (sqft)	Rate per sqft (₹)	Total Boxes
Count	26	26	26
Mean	3497.8	46.84	267.5
Median	1686.8	44.92	94
Std Dev	3893.2	17.62	387.4
Minimum	139.5	26.24	6
Maximum	12400	86.44	1600

Table 2a: Aggregate Descriptive Statistics (All Orders)

The wide range in order quantities (from 139.5 sqft to 12,400 sqft) reflects the company's dual procurement approach: large planned phases and small emergency top-ups. The mean quantity of 3,498 sqft is skewed by several large bulk orders, while the median of 1,687 sqft

better represents a "typical" order. Rate per sqft shows substantial variation (std dev ₹17.62), driven by differences across tile categories rather than price fluctuations for the same model—prices for identical tiles remain stable (e.g., **111 RISER** consistently at ₹27.12).

Tile Type	Count of Orders	Avg Quantity (sqft)	Avg Rate (₹/sqft)
BALCONY	5	5837.5	27.3
FLOOR	8	1793.8	54.7
TERRACE	3	6968.2	49.15
POOL DECK	2	1259.7	49.75
Others*	8	1891.3	49.6

Table 2b: Average Order Metrics by Key Tile Types

*Others include BATHROOM WALL, BATHROOM FLOOR, KITCHEN WALL, RAISER, LIFT FACIA, ROOF, PATHWAY, PLANTER.

Interpretation:

The breakdown by tile type reveals that balcony tiles (**111 RISER**) account for the highest number of orders (5) and the largest average order quantity (5,838 sqft), yet they have the lowest average rate (₹27.3). This combination of high volume and low unit cost makes them a prime candidate for efficiency improvements. Floor tiles, spanning multiple models, exhibit the widest rate variation (₹35.59 to ₹71.31), reflecting the mix of standard and premium design tiles—the catastrophic **CARVED STELLA** (₹69.15) belongs to this premium segment.

The consistent pricing for identical tile models across orders confirms that the problem is not price volatility but operational factors: breakage overrun and design changes. This descriptive overview sets the stage for the detailed case-study analysis that follows.

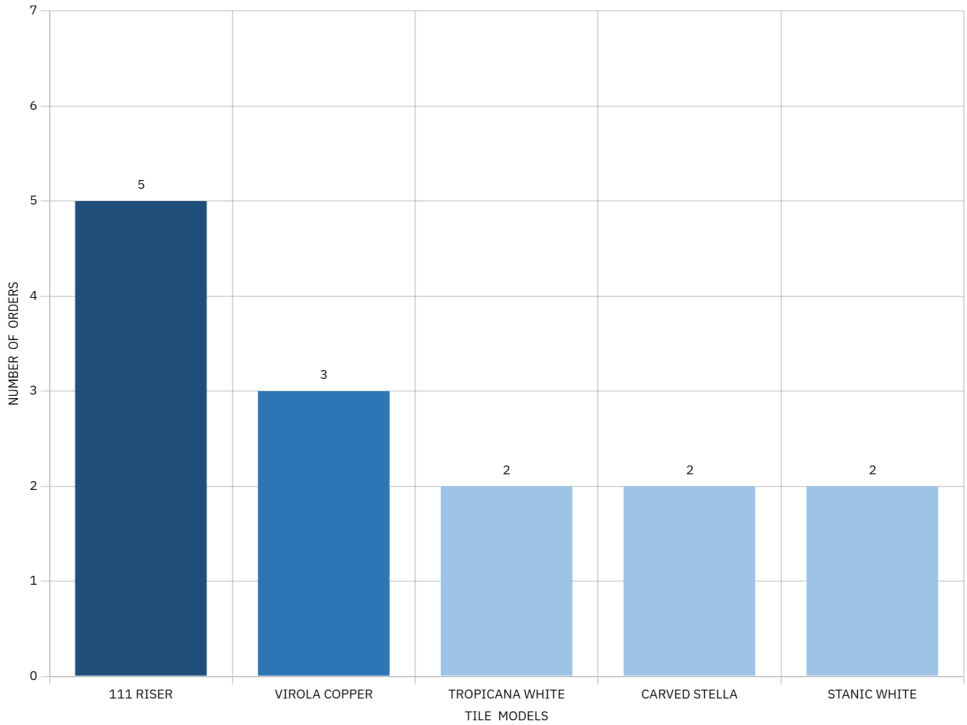


Chart 1: Number of Orders per Tile Model (Top 5 Most Frequent)

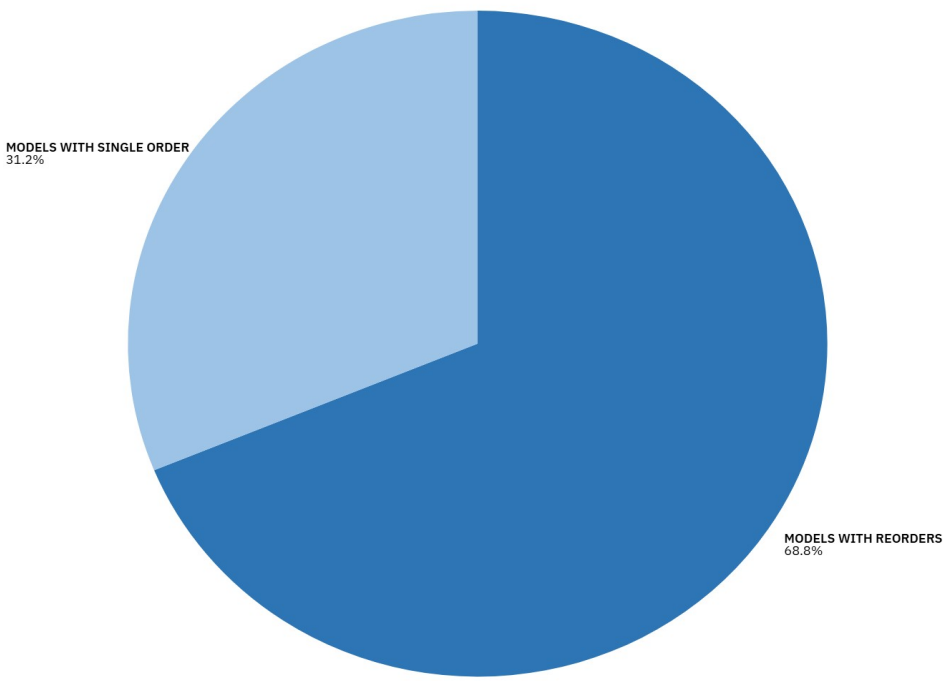


Chart 2: Pie Chart – Proportion of Tile Models with Reorders

6. Analysis Methodology

6.1 Methods Used with Justification

The analysis employed a multi-stage approach:

1. **Cohort Analysis:** Orders were grouped by unique tile model (**Internal Code + Model No**) and arranged chronologically. This allowed tracking of procurement history for each tile type.

2. **Classification Logic:** A two-step rule was applied to distinguish planned phases from emergency shortages:

   - If an order's quantity was significantly smaller (typically <20%) than the immediately preceding order for the same tile, and the time gap was short (weeks rather than months), it was classified as an Emergency order.
   - All other orders were classified as **Planned** (including initial orders and subsequent large orders spaced appropriately).

   Justification: This rule-based approach is appropriate for a small dataset (26 POs) where manual verification is feasible. It aligns with the business reality that phased procurement involves similarly sized orders at regular intervals, while emergency top-ups are small and urgent.

3. **Wastage Calculation:** For each planned phase that was followed by an emergency order, actual wastage was calculated using:

   Actual Wastage % = (**Total Used** – **Planned Need**) / **Planned Need**

   where, **Planned Need** = **Planned Order Quantity** / **1.08** (since the planned order includes the 8% buffer).

   Justification: This formula directly quantifies how much the actual consumption exceeded the planned requirement, isolating the buffer's performance.

4. **Stakeholder Validation:** Preliminary findings were discussed with the HOD to validate interpretations and uncover process-level causes (e.g., design changes) not visible in the data alone. This qualitative step was crucial for accurate root cause diagnosis.

5. **Buffer Simulation:** Different buffer levels (8%, 12%, 15%) were tested against the calculated wastage percentages to determine the minimum buffer that would have prevented each emergency shortage.

Justification: This scenario analysis provides actionable insights by linking buffer adequacy to real-world wastage patterns.

6.2 Analytical Tools Used

The entire analysis was conducted using **Google Sheets** as the primary tool. This choice was deliberate and appropriate for the following reasons:

1. **Dataset Size:** With only 26 purchase orders, the dataset was small enough to be managed efficiently in a spreadsheet environment without requiring programming languages like Python.

2. **Collaborative Validation:** Google Sheets allowed for easy sharing with the HOD during consultation meetings. He could view the structured data, verify entries, and provide real-time feedback on classifications.

3. **Built-in Analytical Features:** The following Google Sheets features were extensively used:
   - **Sorting & Filtering:** To group orders by tile model and arrange chronologically.
   - **Formulas:**
     - =SORT() and =FILTER() for data organization
     - =IF() and =COUNTIF() for tagging reorders

4. **Transparency:** Every step of the analysis—from raw data entry to final calculations—is visible and auditable in the spreadsheet, ensuring complete transparency.

5. **Client Accessibility:** The final deliverable (the analysis workbook with recommendations) can be handed over to Protech Group for ongoing use, as they already use spreadsheet applications in their daily operations.

Note on Python: While the proposal mentioned Python-based time-series forecasting, this approach was deemed unnecessary after understanding that procurement follows discrete project-phase ordering rather than continuous demand. The diagnostic, stakeholder-validated approach using spreadsheets proved more appropriate and actionable.

7. Results and Findings

7.1 Procurement Pattern Overview

Sorting the data by **Internal Code**, **Tile Type**, **Model No**, and **Order Date** revealed that many tiles are purchased in batches of similar size at regular intervals. For example, **TROPICANA WHITE** floor tiles were ordered in 211 boxes and later 800 boxes—both large orders spaced appropriately. This pattern is consistent with **phased construction**, where materials are ordered in stages aligned with project milestones.

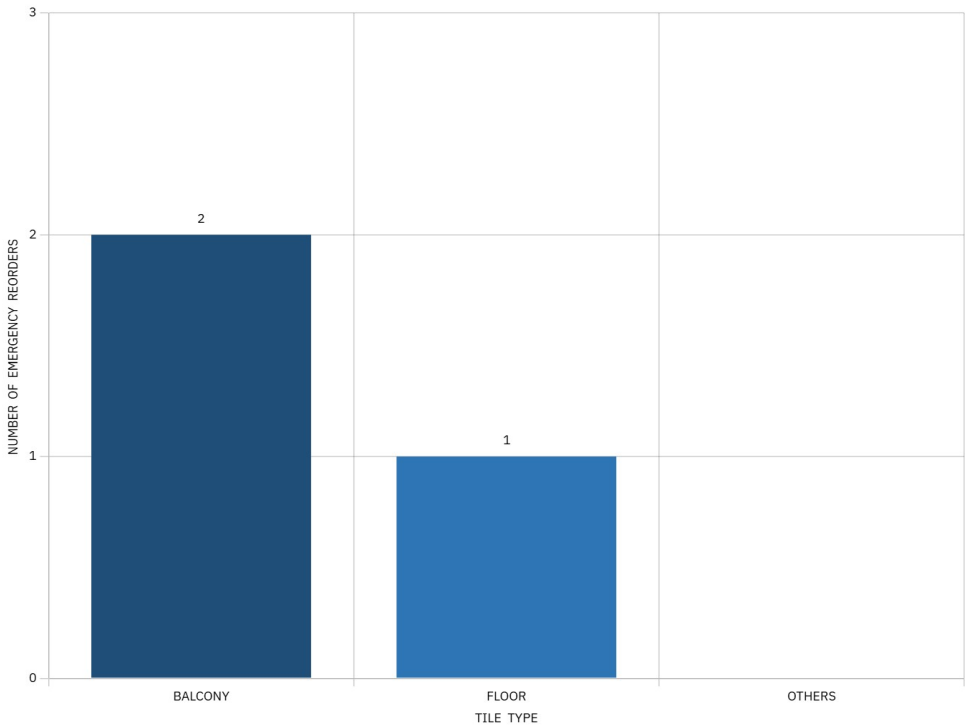


Chart 3: Emergency Reorders by Tile Type

7.2 Distinguishing Planned Phases from Emergency Shortages

Applying the classification logic, two tile models emerged as having genuine emergency shortages: **111 RISER** (balcony) and **CARVED STELLA** (floor).

7.3 Case Study 1: 111 RISER (Balcony Tiles)

This tile model was ordered five times.

Order Sequence	Order Date	Quantity (boxes)	Classification
1	11/12/2024	1590	Planned
2	04/02/2025	59	Emergency
3	24/03/2025	1600	Planned
4	13/06/2025	91	Emergency
5	13/06/2025	426	Planned

Table 3: Order History for 111 RISER

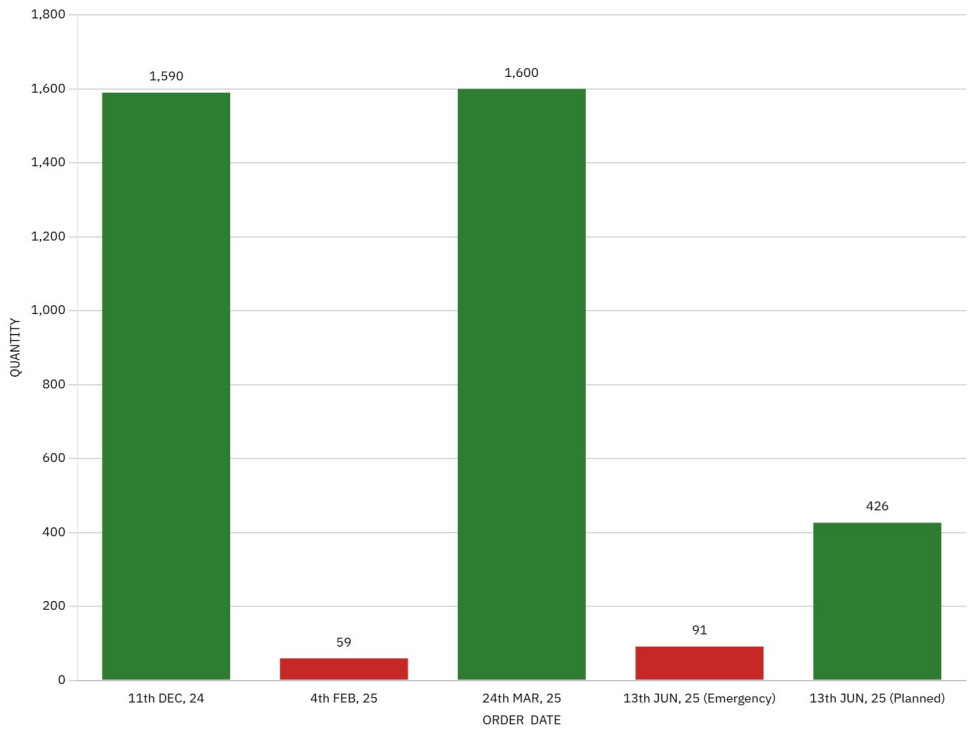


Chart 4: Timeline of 111 RISER Orders

Calculated Wastage:

- **Phase 1 (Seq1 + Seq2):**

  Planned Need = 1590 / 1.08 ≈ 1472 boxes

  Total Used = 1649 boxes

  Actual Wastage = 12.0%

- **Phase 2 (Seq3 + Seq4):**

  Planned Need = 1600 / 1.08 ≈ 1481 boxes

  Total Used = 1691 boxes

  Actual Wastage = 14.2%

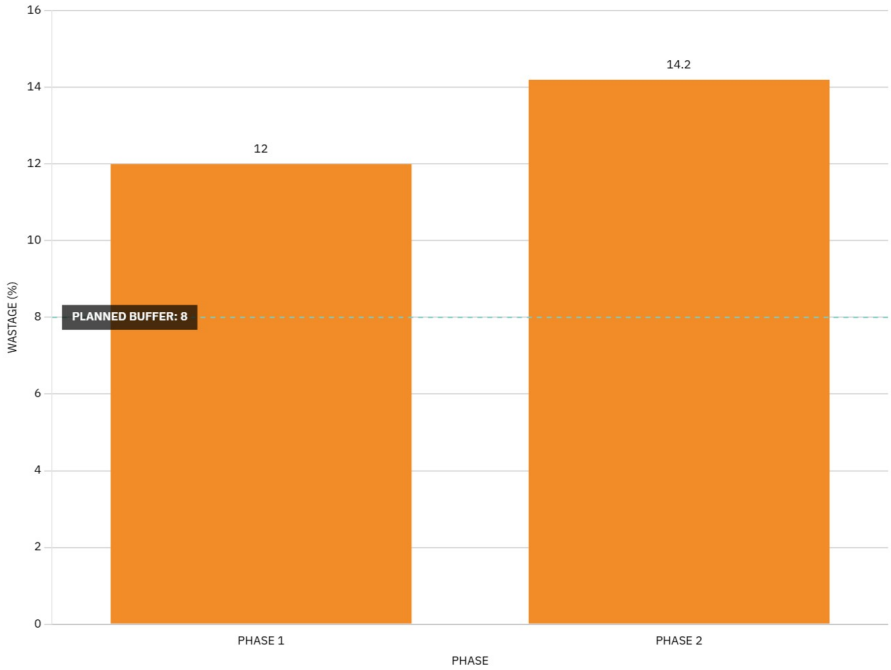


Chart 5: Actual Wastage vs. Planned Buffer for 111 RISER

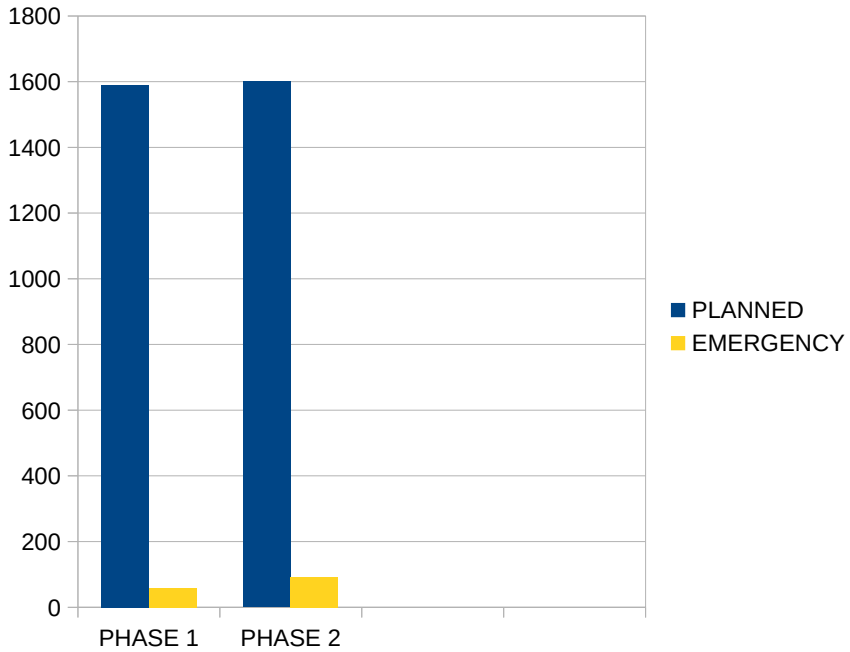


Chart 6: Planned vs. Emergency Order Quantities for 111 RISER

7.4 Case Study 2: CARVED STELLA (Floor Tiles)

This tile model was ordered twice.

Order Sequence	Order Date	Quantity (boxes)	Classification
1	14/05/2025	84	Planned
2	05/06/2025	35	Emergency

Table 4: Order History for CARVED STELLA

Calculated Wastage: 53%

HOD Consultation: Discussion with the HOD revealed the cause: after partial installation, the design was deemed unsatisfactory, and installed tiles were removed and replaced. This is value destruction driven by a design change during construction, not a buffer miscalculation.

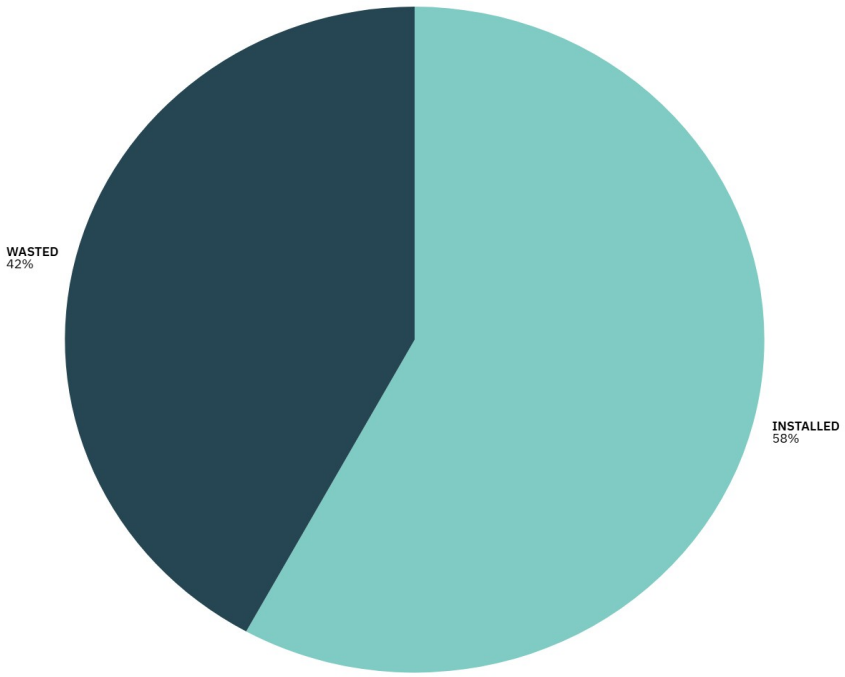


Chart 7: Value Destruction – CARVED STELLA

7.5 Root Cause Allocation

The percentage allocations in Table 5 are informed estimates derived through a combination of quantitative analysis and qualitative validation with the HOD. They are based on the following reasoning:

Event	Unplanned Breakage	Design Volatility	Strategic Reserve
111 RISER Seq2	50%	30%	20%
111 RISER Seq4	50%	30%	20%
111 RISER Seq5	0%	70%	30%
CARVED STELLA Seq2	30%	60%	10%

Table 5: Root Cause Allocation for Emergency Events

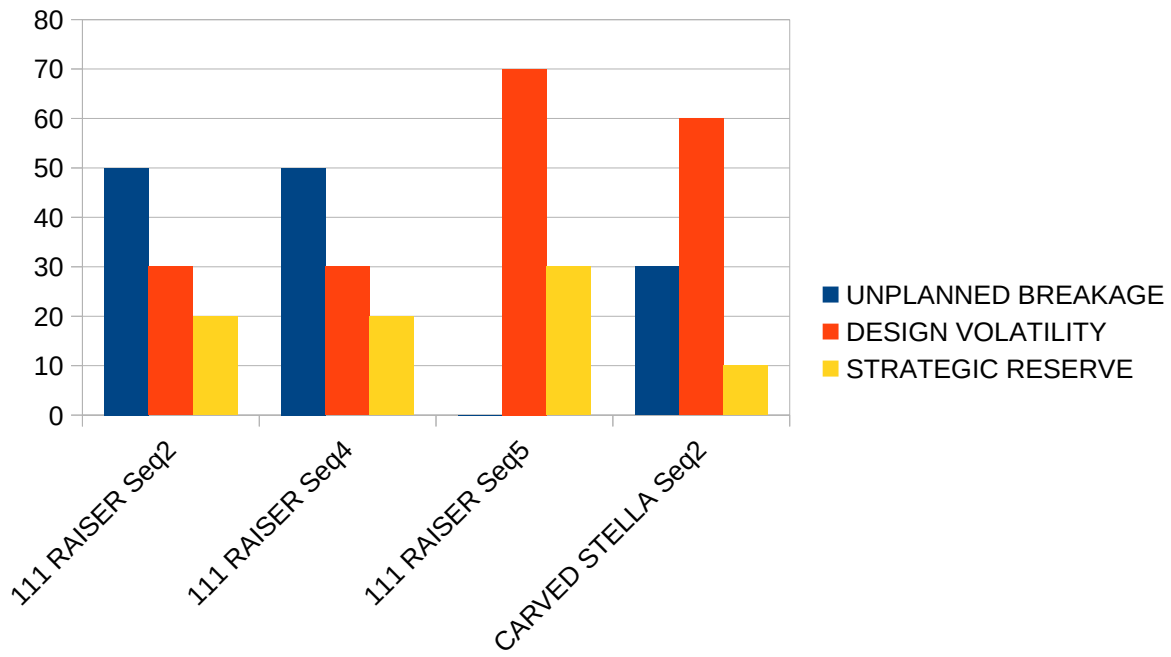


Chart 8: Root Cause Allocation for Emergency Events

7.6 Buffer Simulation and Adequacy Testing

We tested three buffer levels against the observed wastage percentages.

Scenario	Wastage	8% Buffer	12% Buffer	15% Buffer
111 RISER – Phase 1	12%	Fail	Pass	Pass
111 RISER – Phase 2	14.2%	Fail	Fail	Pass
CARVED STELLA	53%	Fail	Fail	Fail

Table 6: Buffer Adequacy Test Results

Key Insight: A 12% buffer would have prevented the Phase-1 emergency but not Phase-2. A **15% buffer would have prevented both emergencies** for **111 RISER**. For **CARVED STELLA**, no practical buffer solves the problem – a process change is required.

8. Interpretation of Results and Recommendations

8.1 Interpretation of Results

The Four Demands on a Single Buffer:

The HOD confirmed that the 8% buffer is intended to cover:

1. Planned breakage (≈5%)
2. Unplanned breakage (variable, often exceeding 5%)
3. Design volatility (post-installation changes)
4. Strategic reserve (sagging, maintenance, handover)

This single buffer is asked to satisfy four distinct, sometimes competing, demands. The analysis shows that for high-risk tiles like **111 RISER**, unplanned breakage alone consumes the entire 8% and more, leaving nothing for the other demands. This explains why emergency reorders occur.

Operational and Management Overhead of Buffer Failure:

When an emergency reorder becomes necessary, the impact extends far beyond the cost of the tiles themselves:

1. **Workflow Disruption and Replanning:** While labor is not left completely idle—workers can be reassigned to other tasks—the planned sequence of activities is disrupted. Site managers must urgently replan the schedule, deciding which tasks can be advanced, which need to be postponed, and how to rearrange the workforce optimally. This creates a cascade of scheduling conflicts that can affect multiple trades.

2. **Management Stress and Overhead:** The most significant cost is the cognitive and time burden on the HOD and site managers. Instead of focusing on strategic planning, quality control, and forward-looking decisions, they are forced into reactive firefighting—coordinating with suppliers, expediting shipments, negotiating with transporters, and constantly updating stakeholders. This stress and diversion of attention is a real but often unquantified cost of buffer failure.

3. **Transportation Uncertainty and Risk:** Emergency reorders are subject to the same long lead times as regular orders, but with additional uncertainty. Small batches may be deprioritized by manufacturers, and freight consolidation becomes challenging. This unpredictability makes it impossible to guarantee when the material will arrive, prolonging the period of uncertainty and disruption.

4. **Value Destruction:** Events like **CARVED STELLA** represent pure value destruction—installed tiles are removed, labor is wasted, and new tiles must be purchased. This doubles the material cost for that area and creates additional waste disposal issues.

While unit tile prices remained stable across reorders, the hidden costs—management overhead, workflow disruption, scheduling uncertainty, and value destruction—are substantial and directly impact project profitability and timeline predictability.

8.2 Recommendations

A universal 8% buffer is ineffective. Operational efficiency requires a **differentiated**, **risk-based procurement strategy** coupled with **process improvements** that address the root causes of buffer failure.

Risk Tier	Category	Example Tile	Primary Wastage Driver	Recommended Buffer	Supporting Action
Tier 1	Standard Tiles	TROPICANA WHITE	Planned Breakage (~5%)	8% (maintain)	None
Tier 2	High Breakage Tiles	111 RISER	Chronic Breakage Overrun (12-14%)	15%	Review Handling/fitting practices; conduct training
Tier 3	Design Volatile Tiles	CARVED STELLA	Post-Installation Design Changes	N/A	Implement **Integrated Design Approval System**

Table 7: Proposed Tiered Buffer Policy

Process Improvements:

1. Integrated Design Approval System:

Instead of relying on a single mock-up for specific tiles, Protech Group should establish a permanent design visualization facility—a dedicated space on-site or at the office where all tile designs intended for a project can be physically laid out before installation begins.

How it works:

- **Step 1 - Graphic Compilation:** For each project, all proposed tile designs are first compiled graphically (digital mood boards, catalog clippings) for initial client/architect shortlisting. This narrows down the options from dozens to a handful.
- **Step 2 - Physical Mock-up Flat:** For the shortlisted designs, a mock-up flat or sample room is created where all selected tiles are actually installed in a representative area. This allows clients and architects to see, touch, and experience the actual tiles in real lighting and space conditions.
- **Step 3 - Formal Approval:** Only after physical inspection and sign-off are the final tile quantities approved for bulk ordering.

Benefits:

- Eliminates post-installation design changes (like the **CARVED STELLA** incident).
- Reduces the number of design iterations and associated wastage.
- Provides a reference standard for installation quality.
- Can be reused across multiple projects with similar design themes.

2. Central Maintenance Stock:

Establish a separate, centrally held inventory of commonly used tiles for post-handover repairs. This stock should be:

- Sourced through a separate annual procurement process.
- Stored in a central warehouse, not at project sites.
- Released only for genuine maintenance needs after project completion.
- Funded by a small allocation from each project's budget (e.g., 0.5% of project cost).

This prevents project-specific buffers from being drained for post-handover requirements.

3. Handling Practice Review and Training:

For tiles identified as high-breakage (Tier 2):

- Conduct a **video-based handling audit:** Record unloading, shifting, and installation processes for 2-3 days.
- Review the footage with site supervisors to identify specific practices that may be causing breakage.
- Develop a **standard operating procedure (SOP)** for handling fragile/large-format tiles.
- Conduct quarterly training sessions for all tiling crews.

4. Supplier Collaboration and "Buffer Banking":

For frequently reordered tiles (like **111 RISER**), negotiate with key suppliers to maintain a **supplier-held buffer stock**. The company commits to purchasing a certain minimum annual volume, and the supplier agrees to keep a small quantity (e.g., 50 boxes) readily available at their Guwahati dealership for emergency dispatch. This reduces lead time from 30 days to 2-3 days for emergency needs.

5. **Procurement Dashboard and Early Warning System:**

Develop a simple Excel-based dashboard that:

- Tracks actual vs. planned buffer consumption for each tile type.
- Automatically flags tiles where usage exceeds 80% of the buffer, alerting the procurement team to review site progress and anticipate potential shortages.
- Maintains a history of wastage rates by tile model to inform future buffer calculations.

6. **Post-Project Review and Knowledge Base:**

After each project, conduct a **post-project review** specifically focused on tile procurement and wastage. Document:

- Which tiles exceeded buffers and why.
- Any design changes that occurred and their causes.
- Supplier performance (delivery times, quality).
- Lessons learned and updated buffer recommendations for similar future projects.

Maintain this as a **living knowledge base** in a shared spreadsheet or simple database. This transforms anecdotal learning into institutional memory.

7. **Tie-Breaker Matrix for Supplier Selection:**

During supplier selection, use a weighted decision matrix that includes not just price but also:

- Lead time reliability (historical on-time delivery %)
- Responsiveness to small emergency orders
- Distance from site (affects freight cost and time)
- Willingness to hold buffer stock

This ensures that the chosen supplier aligns with the company's need for both cost-efficiency and operational flexibility.

8. **Monthly Procurement Health Meeting:**

Institutionalize a monthly 30-minute meeting between the HOD, site supervisors, and the procurement team to review:

- Current buffer status for all active tiles.
- Any emerging design changes or client requests.
- Supplier performance issues.
- Upcoming project phases and anticipated material needs.

This shifts the focus from reactive firefighting to proactive planning.

8.3 Implementation Roadmap

The recommendations are designed to be implemented in phases, respecting the company's existing workflows and resource constraints.

Phase 1: Immediate Actions (Next Project – Within 1 Month)

Action	Responsibility	Success Metric
Pilot 15% buffer on all balcony tiles (like **111 RISER**)	Procurement Team	Track emergency reorders for these tiles; compare to historical data
Create design visualization facility – allocate a small room/space for physical mock-ups	HOD / Site Manager	Mock-up space ready before next design phase; at least one design finalized using the facility
Compile graphic catalog of all tile designs for client shortlisting	Design/Architecture Team	Client feedback on ease of selection; reduction in design iterations
Initiate supplier discussion on "buffer banking" for frequently reordered tiles	HOD / Procurement	At least one supplier agrees to pilot buffer banking for a specific tile
Build basic procurement dashboard in Excel (early warning system)	Procurement	Dashboard functional and handed over to procurement team

Phase 2: Short-term Actions (3 Months)

Action	Responsibility	Success Metric
Develop handling SOPs for fragile/large-format tiles	Site Supervisors	SOP document created and shared with all site teams
Conduct first training session for tiling crews	HOD / External Trainer	Training attendance ≥80% of crews; feedback score ≥4/5
Establish central maintenance stock – identify budget and storage location	HOD / Finance	Stock physically procured and stored; clear release protocol documented
Formalize monthly procurement health meeting	HOD	Meeting held for 3 consecutive months with documented minutes and action items
Implement post-project review for current/next project	HOD / Site Team	Review conducted; findings documented in knowledge base

Phase 3: Long-term Institutionalization (6-12 Months)

Action	Responsibility	Success Metric
Review pilot results – analyze impact of 15% buffer on balcony tiles	HOD	Quantitative report on reorder reduction; cost-benefit analysis
Adjust buffer tiers based on collected data	HOD / Procurement	Updated tier policy documented and communicated
Expand design visualization facility to permanent setup if successful	Management	Facility used for all new projects; client satisfaction with design process tracked
Integrate learnings into procurement policy – update standard operating procedures	HOD	Revised procurement manual approved and distributed
Share findings with other project teams within Protech Group	HOD	Cross-team knowledge sharing session conducted
Evaluate supplier buffer banking performance and expand to more tiles/suppliers	Procurement	At least 2 more tiles covered under buffer banking arrangements

KPIs to Track:

Metric	Baseline	Target (After 1 Year)
Number of emergency reorders per project	[Insert baseline from this analysis]	Reduce by 50%
Average buffer overrun for high-risk tiles	12-14%	≤ 5% overrun
Design change-related reorders	2 events	Zero
Management time spent on procurement firefighting (estimated)	High	Reduced by 50% (measured via self-reporting)
Client satisfaction with design finalization	Not tracked	Measured via feedback; target ≥4/5
Supplier lead time for emergency orders	30+ days	≤ 7 days for buffer-banked items

9. Conclusion

This project analyzed 26 primary purchase orders from Protech Group to diagnose the root causes of tile procurement inefficiency. The analysis revealed that the standard 8% buffer fails not because it is too small, but because it is undifferentiated—it must cover breakage, design volatility, and maintenance simultaneously. For high-risk tiles like **111 RISER**, actual wastage of 12-14% consistently exceeds the buffer, forcing emergency reorders. For design-critical tiles like **CARVED STELLA**, a 53% reorder was driven by post-installation design changes, a process failure beyond inventory control.

The true cost of these failures extends beyond tile prices. It includes management overhead, workflow disruption, scheduling uncertainty, and value destruction—all of which erode project profitability and predictability.

The recommended integrated solution—a tiered buffer policy (15% for high-breakage tiles), an integrated design approval system with physical mock-up facilities, central maintenance stock, supplier buffer banking, a procurement dashboard, and regular health meetings—directly addresses these distinct pressures. It preserves the company's efficient phased procurement practice while adding resilience where it is most needed.

By implementing these recommendations, Protech Group can reduce emergency reorders, minimize management firefighting, enhance operational efficiency, and deliver more predictable, profitable projects. The analysis and tools developed during this project provide a strong foundation for this transformation.